

Windows 11 Native Laravel 13 Development Setup

Apache 2.4 + PHP 8.4 + MySQL 8.4 + Composer + Node.js + VS Code Xdebug

Prepared for: A Rajashekhar Reddy | Paths use capital F: drive notation

1. Final Working Setup Summary

Component	Version / Status	Important Path / URL
OS	Windows 11	Native setup, no Docker
Apache	Apache HTTP Server 2.4	F:\devtools\apache24
PHP	PHP 8.4.22 Thread Safe x64	F:\devtools\php84
MySQL	MySQL Community Server 8.4 LTS	Data: F:\databases\mysql\Data
Composer	Composer 2.10.1	Uses F:\devtools\php84\php.exe
Node.js / npm	Node v24.15.0 / npm 11.12.1	Installed globally
Laravel	Laravel 13 project	F:\devtools\apache24\htdocs\medisys
Local domain	Apache virtual host	http://medisyslocal.com
Debugging	Xdebug + VS Code working	Port 9003

2. Official Download Links Used

- Visual C++ Redistributable x64: https://aka.ms/vs/17/release/vc_redist.x64.exe
- Apache Lounge Apache 2.4 Windows builds: <https://www.apachelounge.com/download/>
- PHP for Windows 8.4 downloads: <https://windows.php.net/download/>
- MySQL Community Server 8.4 LTS: <https://dev.mysql.com/downloads/mysql/8.4.html>
- Composer Windows installer: <https://getcomposer.org/download/>
- Node.js LTS download: <https://nodejs.org/en/download>
- VS Code download: <https://code.visualstudio.com/Download>
- Xdebug Wizard: <https://xdebug.org/wizard>

3. Folder Structure on F: Drive

Folders created with lowercase names, while path examples use capital F: as requested.

F:\devtools
F:\devtools\apache24
F:\devtools\php84
F:\devtools\vscode
F:\databases\mysql
F:\projects\laravel
F:\backups
F:\downloads

4. Apache 2.4 and PHP 8.4 Setup

1. **Extract Apache:** Extract Apache Lounge ZIP to F:\devtools\apache24.
2. **Fix ServerRoot:** In F:\devtools\apache24\conf\httpd.conf, set Define SRVROOT "F:/devtools/apache24" and ServerRoot "\${SRVROOT}".
3. **Extract PHP:** Extract PHP 8.4 x64 Thread Safe ZIP to F:\devtools\php84 and rename php.ini-development to php.ini.
4. **Environment PATH:** Add F:\devtools\php84 and F:\devtools\apache24\bin to Windows PATH.

4.1 PHP php.ini Changes

The extension_dir issue was fixed by explicitly pointing PHP to the F: drive extension folder:

```
extension_dir = "F:\devtools\php84\ext"
```

Enabled extensions:

```
extension=curl
extension=fileinfo
extension=gd
extension=intl
extension=mbstring
extension=mysqli
extension=openssl
extension=pdo_mysql
extension=sodium
extension=zip

memory_limit=512M
upload_max_filesize=64M
post_max_size=64M
max_execution_time=120
date.timezone=Asia/Kolkata
```

4.2 Apache PHP Module Configuration

Add the following to httpd.conf:

```
LoadModule php_module "F:/devtools/php84/php8apache2_4.dll"
PHPIniDir "F:/devtools/php84"
AddHandler application/x-httpd-php .php
DirectoryIndex index.php index.html

LoadModule rewrite_module modules/mod_rewrite.so
ServerName localhost:80
```

4.3 Install and Test Apache Service

Run Command Prompt as Administrator:

```
cd F:\devtools\apache24\bin
httpd -t
httpd -k install
httpd -k start
```

Verify Apache and PHP:

http://localhost
http://localhost/info.php

5. MySQL 8.4 LTS Setup

MySQL binaries were installed by the installer under C:\Program Files, but the important database data directory was set to F: drive.

Program files:
C:\Program Files\MySQL\MySQL Server 8.4\bin

Data directory:
F:\databases\mysql\Data\

Add this MySQL bin path to Windows PATH:

C:\Program Files\MySQL\MySQL Server 8.4\bin

Verify MySQL:

```
mysql --version
mysql -u root -p
SHOW VARIABLES LIKE 'datadir';
```

Create Laravel database:

```
CREATE DATABASE medisys CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
SHOW DATABASES;
```

6. Composer Setup and PHP 8.4 Fix

Composer was previously pointing to WinGet PHP 8.3. After removing PHP 8.3, composer.bat was updated to use PHP 8.4.

composer.bat final content:

```
@echo off
"F:\devtools\php84\php.exe" "C:\Users\araja\bin\composer.phar" %*
```

Verify:

```
composer self-update
composer -V
composer diagnose
```

Expected: Composer 2.10.1 and PHP 8.4.22 using F:\devtools\php84\php.exe.

7. Laravel 13 Project Setup

Install Laravel installer globally:

```
composer global require laravel/installer
```

If laravel command is not recognized, add Composer global vendor bin to PATH:

C:\Users\araja\AppData\Roaming\Composer\vendor\bin

Project location used:

```
F:\devtools\apache24\htdocs\medisys
```

Useful commands:

```
cd F:\devtools\apache24\htdocs\medisys
php artisan --version
php artisan key:generate
php artisan migrate
```

7.1 Laravel .env Configuration

Use native Windows MySQL host, not Docker host mysql:

```
APP_URL=http://medisyslocal.com

DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_PORT=3306
DB_DATABASE=medisys
DB_USERNAME=root
DB_PASSWORD=your_mysql_password
```

If artisan fails after .env changes, clear cache:

```
php artisan optimize:clear
Remove-Item bootstrap\cache\*.php
```

8. Apache Virtual Host for Laravel

Using the public folder is correct and production-like. The web server should point to the Laravel public directory, not to the project root.

Enable vhosts in httpd.conf:

```
Include conf/extra/httpd-vhosts.conf
```

Add this to F:\devtools\apache24\conf\extra\httpd-vhosts.conf:

```
<VirtualHost *:80>
    ServerName medisyslocal.com
    DocumentRoot "F:/devtools/apache24/htdocs/medisys/public"

    <Directory "F:/devtools/apache24/htdocs/medisys/public">
        Options Indexes FollowSymLinks
        AllowOverride All
        Require all granted
    </Directory>
</VirtualHost>
```

Add this line at the bottom of hosts file:

```
File:
C:\Windows\System32\drivers\etc\hosts
```

Line:
127.0.0.1 medisyslocal.com

Restart Apache and open the site:

```
cd F:\devtools\apache24\bin  
httpd -k restart
```

<http://medisyslocal.com>
<http://medisyslocal.com/dashboard>

9. Node.js and Laravel Frontend Assets

Installed versions: Node.js v24.15.0 and npm 11.12.1.

Verify and build assets:

```
node -v  
npm -v
```

```
cd F:\devtools\apache24\htdocs\medisys  
npm install  
npm run build
```

```
# for active development  
npm run dev
```

10. Xdebug + VS Code Setup

Xdebug was installed and VS Code debugger is working.

Generate PHP info for the Xdebug Wizard:

```
php -i > F:\downloads\phpinfo.txt
```

Use Xdebug Wizard to select the correct DLL, then copy it to:

```
F:\devtools\php84\ext\php_xdebug.dll
```

Add to bottom of F:\devtools\php84\php.ini:

```
zend_extension=xdebug  
  
xdebug.mode=debug,develop  
xdebug.start_with_request=yes  
xdebug.client_host=127.0.0.1  
xdebug.client_port=9003  
xdebug.log="F:\devtools\php84\xdebug.log"
```

Restart Apache and verify:

```
cd F:\devtools\apache24\bin  
httpd -k restart
```

```
php -v  
php -m | findstr xdebug
```

10.1 VS Code launch.json

Install VS Code extension: PHP Debug. Then create .vscode\launch.json:

```
{
  "version": "0.2.0",
  "configurations": [
    {
      "name": "Listen for Xdebug",
      "type": "php",
      "request": "launch",
      "port": 9003
    }
  ]
}
```

11. Issues Fixed During Setup

- **PHP extensions tried loading from C:\php\ext:** Fixed by setting extension_dir = "F:\devtools\php84\ext" in php.ini.
- **Apache ServerRoot error:** Fixed by changing SRVROOT to F:/devtools/apache24.
- **ServerName warning:** Fixed by setting ServerName localhost:80.
- **Laravel DB host mysql failed:** Fixed by changing DB_HOST from mysql to 127.0.0.1.
- **Apache route Not Found for /dashboard:** Fixed with VirtualHost, AllowOverride All, and mod_rewrite.
- **Composer using PHP 8.3:** Fixed by removing WinGet PHP 8.3 and editing composer.bat to use F:\devtools\php84\php.exe.
- **Invalid URI: Scheme is malformed:** Fixed by correcting APP_URL to http://medisyslocal.com and clearing Laravel cache.

12. Final Verification Commands

Run these commands to verify the complete setup:

```
php -v
php -m
httpd -v
httpd -M | findstr rewrite
mysql --version
mysql -u root -p
composer -V
composer diagnose
node -v
npm -v

cd F:\devtools\apache24\htdocs\medisys
php artisan --version
php artisan migrate
php artisan route:list
npm run build
```

13. Pending Components for Later

- Redis / Memurai
- MongoDB Community Server
- MongoDB PHP extension
- SQL Server 2019 Developer
- SSMS
- SQL Server PHP drivers: sqlsrv and pdo_sqlsrv

End of setup document